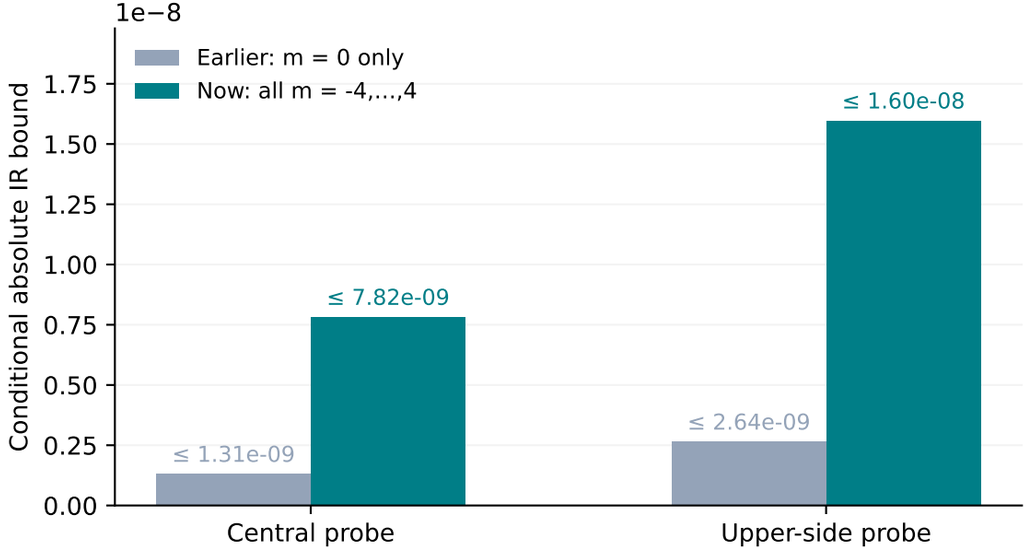
# More infrared sectors covered; continuum work remains

Weighted energy + radial-pressure source · Selected phase π · Prescribed geometry, not a physical medium

Assumed long throat
ρc = 32; b = 0.2d

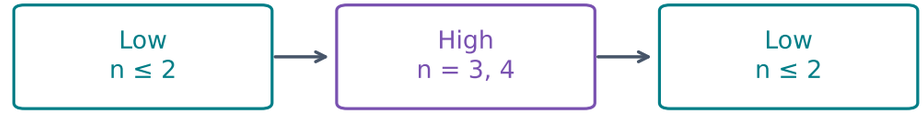
Sharp junction source and
physical preparation unresolved

## A. IR: 1 → 9 retained azimuthal sectors




$0 \le x=\kappa\ell \le 10^{-6}$; finite N4/J9/DG, floating conditional bounds.
More sectors covered; this is still not the infinite angular sum.

## B. Finite angular-feedback benchmark




Mode coupling in saved matrices; not a travel path.

| | |
|---|---|
| Central probe | **\|feedback\| ≤ 4.65e-04** |
| Upper-side probe | **\|feedback\| ≤ 1.79e-04** |

At x = 0.4439; pointwise per dκ.
Finite N2→N4 only; no infinite-angular or integral bound.

**Illustrative continuum requirement — not a measured residual**

For a 1% sampled-kernel feedback budget: right/dual L²-Frobenius residual norms each ≤ 2.25e-05 central / 1.39e-05 upper-side.

Assumes full trial high-mode traces are controlled by the finite norms. Those continuum fields and residuals are not yet available.

Exact angular-quadrature truncation is bounded; floating arithmetic and continuum discretization are not certified by that result.

No combined full-source bound, supporting geometry or stability follows from these separate checks. Units: $\hbar = c = d = 1$.

**No stable medium or 13 Sep 2027 → 12 Sep 2027 passage demonstrated.**